



Meta-analysis on the influence of AI agents on K-12 student cognitive performance

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ABSTRACT

Background: Following rapid evolution of AI in education, AI agents are being integrated into K-12 classrooms. However, their effectiveness on student cognitive enhancement remains inconsistent.

Objectives: This study conducted a comprehensive meta-analysis to evaluate the effectiveness of AI agents on student cognitive learning outcomes in K-12 education, and their influence on specific cognitive categories, while highlighting the influence of moderator variables, such as type of AI agents, learner level, discipline, and experimental duration.

Methods: A random-effect meta-analytic approach was employed, synthesizing 73 effect sizes from 34 studies published between 2020 and 2025. The main effect size was calculated using Hedge's g , while the heterogeneity was assessed using the Q , I^2 , and τ^2 statistics. Publication bias was tested using the funnel plot, while the Classic fail-safe N test, Orwin's fail-safe N test, and trim-and-fill methods were used to assess the robustness of findings.

Results and conclusion: Results demonstrate that: 1) AI agents have a statistically significant, moderate positive effect on student cognitive learning outcomes (with an overall effect size of 0.404, $p < 0.001$); 2) Across cognitive categories, AI agents showed a moderate significant influence on skills-based cognitive outcomes ($g = 0.391$, $p < 0.001$) and knowledge-based cognitive outcomes ($g = 0.344$, $p < 0.05$); 3) For higher-order thinking skills, the influence was insignificant ($g = 0.540$, $p = 0.066$); 4) There was a significant moderating effect of learner level and discipline; 5) The publication bias test showed a slight study effect, whereas sensitivity analysis confirmed robustness of results.

Implications: Findings indicate the need to consider both the disciplinary context and students' characteristics when integrating AI agents in K-12 classrooms. The results provide direction to educators, policymakers, and researchers on the effectiveness of AI agents on student cognitive learning outcomes in K-12 education.

1. Introduction

A potentially positive impact of artificial intelligence (AI) in education has been widely hypothesized (European Education and Culture Executive Agency, 2025). Following broad advancement of AI technology, research on its influence on education is increasing, and several categories of learning outcomes, including cognitive outcomes, are being assessed. To this end, AI agents have been found to have notable applications, such as coaching, tutoring, assessment, and feedback, with the potential to improve students' knowledge acquisition, skills development, and higher-order thinking skills (Gillani et al., 2023).

Since adoption of AI technology have been widely proposed, integration of AI in K-12 classrooms has been increasing since 2019

(Marques et al., 2020). K-12 education is rapidly becoming a promising field in the development of AI-related competency (Sanusi et al., 2022). AI applications at K-12 level aims to develop core skills needed for the future, as primary and secondary education are crucial for developing basic and higher-order thinking skills (Wang et al., 2024). Several studies have suggested that, when integrated into K-12 classrooms, AI agents support personalized learning, adaptive scaffolding, and immediate feedback, which are essential elements for developing student cognitive learning outcomes (Alneyadi & Wardat, 2023; Haelermans & Ghysels, 2017; Kuhlmann et al., 2024). For instance, Wongwatkit and Panjaburee (2024) in their experimental study, they report that AI-based intelligent tutoring systems enhanced lower secondary school students' digital literacy skills by providing guidance and adaptive clues. Similarly,

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research on generative AI (GenAI), conversational AI, and AI robotics, has shown that interactive support and immediate feedback have contributed to the development of middle school students' higher-order thinking skills (Istrate, 2024).

Understanding the influence of AI agents in K-12 education requires not only examining their overall effectiveness but also systematically investigating factors that may influence it, such as learners' educational level, the discipline under implementation, intervention duration, and the instructional strategy used (Díaz & Nussbaum, 2024; Liu & Zhong, 2024; Velander et al., 2024).

Despite a growing body of literature on AI technology in K-12 education, the empirical evidence is surprisingly mixed. While some studies report positive results, such as improvements in students' competencies and skills across disciplines (Huang & Qiao, 2024; Khazanchi et al., 2025; Tan et al., 2025; Ye et al., 2025), others report negative or mixed results on students' learning outcomes (Maamor et al., 2025; Richter et al., 2019; Vieriu & Petrea, 2025). These inconsistent findings may arise from how cognitive learning domains have been theorized and synthesized in previous research. Recent literature reviews and syntheses (e.g., Liu et al., 2025) mainly aim to summarize the overall influence of AI technology in K-12 education, but lack a specific, detailed synthesis of the effectiveness of AI agents in facilitating student cognitive learning outcomes and how these effects may vary due to different moderating factors. As a result, a key question remains as to how AI agents differentially impact student cognitive outcomes, such as knowledge acquisition, problem-solving skills, and higher-order thinking. Therefore, to examine whether AI technology is conducive to student cognitive development in K-12 settings, this study employed a meta-analysis by synthesizing 34 (quasi-) experimental studies published between January 2020 and December 2025. To account for the variability of the included studies, a random-effects model was used to estimate the effect sizes. Subgroup and moderator analyses were conducted to examine the potential sources of heterogeneity, specifically addressing the following research questions:

- 1) How effective are AI agents in facilitating K-12 student cognitive learning outcomes?
- 2) What is the effectiveness of AI agents across cognitive learning categories?
- 3) How do research factors such as type of AI agent, learner level, discipline, and experimental duration influence potential effects?

2. Relevant literature review

2.1. AI agents in education

The concept of AI dates back to the early 1950s, when Alan Turing proposed that a machine could be regarded as "intelligent" if it could imitate human behavior successfully, making it indistinguishable from that of a human (Russell & Norvig, 2016). Following Turing's work, different types of AI technology have been developed (e.g., intelligent tutoring systems, conversational AI, GenAI, AI-educational robotics, etc.) (Díaz & Nussbaum, 2024; Tan et al., 2025). The potential of AI technology as an educational intervention revolution is being extensively examined at all levels of education. Previous research shows that AI agents can facilitate student cognitive learning by facilitating engagement, personalized learning, and problem-solving abilities (Ali et al., 2021; Khazanchi et al., 2025; Tasdelen & Bodemer, 2025).

Several definitions and applications of AI agents in educational contexts have been identified (Akgun & Greenhow, 2022). Recent definition by Istrate (2025) conceptualizes AI agents as software tools that perform exactly the roles as humans and can engage in human-like activities such as instructing, adapting, problematizing, and learning. Consequently, this study defines AI agents as AI-driven systems that interact with learners through instructional, conversational, and adaptive approaches to support the development of cognitive learning

outcomes in K-12 students.

2.2. AI and cognitive learning outcomes in K-12 education

In many cases, scholars categorize cognitive skills into lower-level skills such as knowledge acquisition and comprehension, as well as higher-level skills like problem-solving and creativity (Wei et al., 2021). These particular skills are considered essential to prepare students for higher education and future careers (UNESCO, 2025). Recently, many researchers have conducted extensive studies on the relationship between AI in instruction and cognitive learning outcomes among K-12 students ((Lee & Kwon, 2024). Prior literature indicates that AI agents play a crucial role in shaping student cognitive development by providing learners with opportunities to practice, receive feedback, and reflect on their learning processes. For instance, a study by Fang et al. (2025) conducted a quasi-experimental study with 80 high school students divided into an experimental group (using the Reflective GenAI Debugging approach) and a control group (using the Conventional Coding and Debugging approach) to evaluate the influence of AI-integrated methods on students' computational thinking. The results showed that the AI-integrated approach enhanced students' development of computational thinking skills more than the conventional method. Similarly, Baha et al. (2023) investigated the effectiveness of educational chatbots with 109 upper secondary school students and found that chatbots enhance students' learning experience by allowing them to learn at their own pace, keep them engaged, and save time.

However, there is inconsistency in findings. Some studies argue that AI agents do not significantly support the development of student cognitive abilities. For instance, Tong et al. (2025) compared the effects of human-AI collaboration and human-human collaboration on student cognitive outcomes. They found that effect size for the human-AI collaboration group was notably smaller than that of the control group that did not use AI. This is consistent with Yang et al. (2025), who report that ChatGPT's effectiveness in facilitating the learning achievement of high school students did not meet expectations, as students using ChatGPT had lower achievement compared to those utilizing traditional methods.

2.3. Reviews of the previous studies related to the current study

Following the rapid growth of AI technology in education worldwide, a substantial number of review studies have been conducted to summarize and synthesize its effect on teaching and learning. Appendix A summarizes 11 literature reviews published recently (2024-2025). Despite the variety of literature reviews, the main focus has been to provide an overview of the effects of AI technology in K-12 education. Most of the reviews conducted focus on the broader learning outcomes, without specifically evaluating the effectiveness of AI technology on cognitive learning outcomes in the K-12 context (Azzam & Charles, 2024; Deng et al., 2025; Istrate, 2025; Lee & Kwon, 2024; Liu & Zhong, 2024; Palioura & Sapounidis, 2025; Zhu et al., 2025). Only a study by Yi et al. (2025) goes specifically to examine the influence of AI on students' mathematics performance in K-12. However, the study only focuses on general AI, which lacks a critical discussion on how specific AI agents, such as AI-intelligent tutoring systems, GenAI, and large language models, are shaping student cognitive learning outcomes. More recently, Wang and Fan (2025) conducted a meta-analysis on the effect of ChatGPT as a single tool on students' learning perceptions, performance, and higher-order thinking. The authors reported a large positive impact on learning performance ($g = 0.867$), and a moderate effect on both higher-order thinking ($g = 0.457$) and learning perception ($g = 0.456$); however, their outcomes combine learning performance, learning perception, and higher-order thinking, and the study was not situated in the K-12 context. Consequently, while the existing reviews have evaluated the effectiveness of AI technology in students' learning outcomes, only a few have considered the influence of moderator variables such as

learners' education level, discipline under AI intervention, and duration of intervention (Liu, Guo, & Hu, 2025; Palioura & Sapounidis, 2025).

Therefore, due to inconsistencies on the relationship and direction of AI's influence on student cognitive development, as well as the gaps from previous studies that do not address cognitive learning domains, the current study quantitatively synthesizes the effect of AI agents on student cognitive learning outcomes in K-12 education. The study further highlights how this particular effect is influenced by moderator variables, such as the type of AI agent, learners' level, discipline, and experimental duration, in different dimensions of cognitive categories, including knowledge-based outcomes, skills-based outcomes, and higher-order thinking.

3. Method

3.1. Data sources and searching strategy

To obtain the relevant literature, a systematic search was conducted across several databases, including Web of Science, ScienceDirect, Google Scholar, and Springer Nature. A citation approach was employed to retrieve the literature, ensuring comprehensive coverage. The literature search was conducted for the empirical studies published from January 2020 to December 2025.

The key terms used in the literature search combined keywords related to AI agents and students' learning outcomes in K-12 classrooms, including "AI agent" AND "learning" AND ("K-12" OR "primary school" OR "secondary school" OR "high school" OR "elementary school"); "AI agents" AND "students learning" AND ("K-12" OR "primary school" OR

"secondary school" OR "high school" OR "elementary school"); "AI agents" AND "education" AND ("K-12" OR "primary school" OR "secondary school" OR "high school" OR "elementary school"); "AI agents" AND "cognitive outcomes" AND "K-12".

In addition, the study looked for the synonyms of some terms, such as learning, cognitive, and outcome, as supplementary search terms in the search process to expand the scope of the literature search.

3.2. Search results

The study selection process adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guideline. The corresponding flow of study identification, screening, eligibility, and inclusion is illustrated in Fig. 1. Subsections 3.2.1 to 3.2.4 describe each screening segment in detail, aligned with the PRISMA flowchart.

3.2.1. Initial screening

During the PRISMA screening phase, literature that did not serve the purpose for this meta-analysis were removed, leaving only empirical studies relevant for examining the influence of AI agents on student cognitive learning outcomes in K-12 education. The inclusion criteria for literature screening were as follows: first, the study is on the influence of AI agents on student cognitive learning outcomes in K-12 settings; second, the study is an experimental or quasi-experimental design; third, the main participants in research are students and not teachers or any other objects; fourth, the study data is complete, including the primary data such as the mean test scores in both the experimental and control groups, and the number of participants in the two groups, essential for

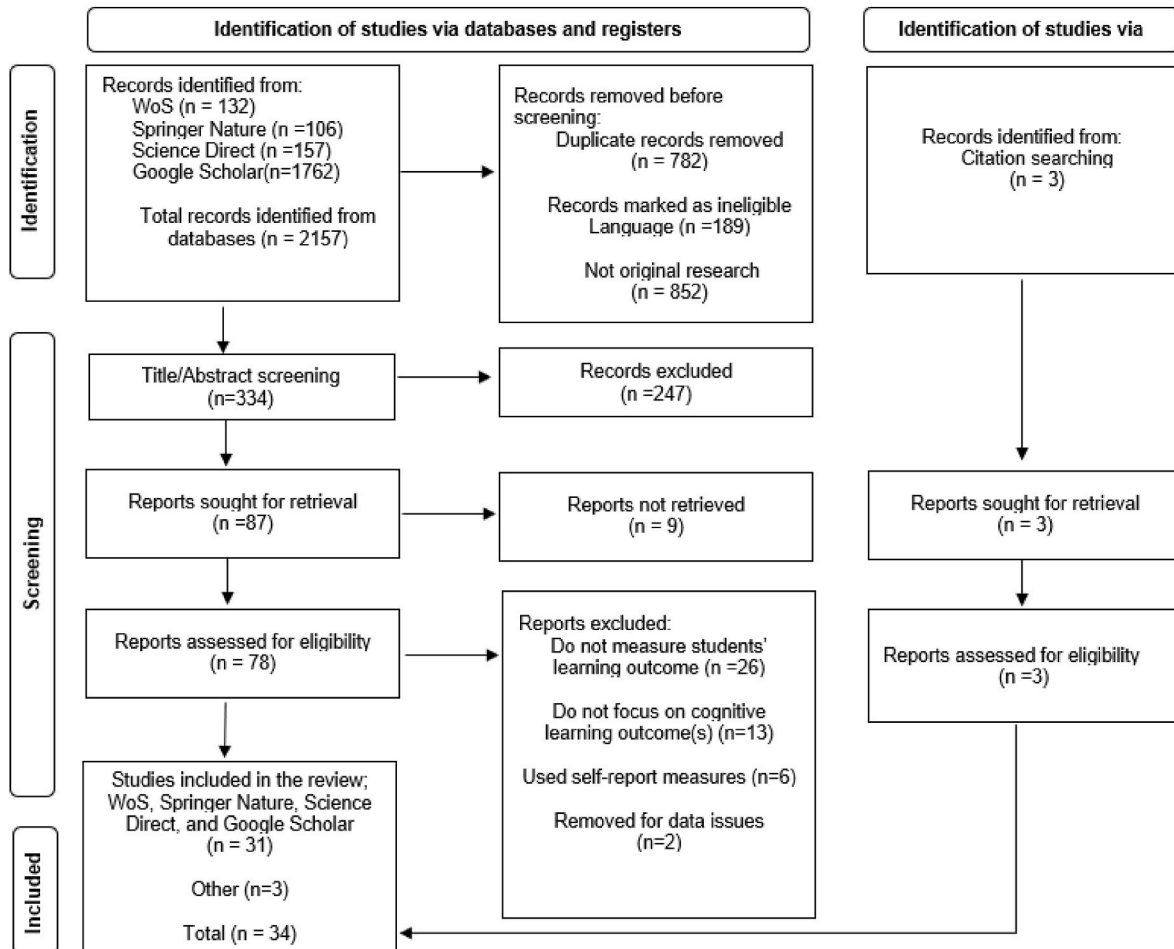


Fig. 1. The PRISMA flow diagram.

calculating the effect sizes; and fifth, the study is published in English.

A total of 2157 articles were initially obtained based on the criteria used in the searching procedures (132 from Web of Science, 106 from Springer Nature, 157 from ScienceDirect, and 1762 from Google Scholar), as well as 3 articles through the citation method. From these results, 782 duplicates were removed, 189 publications were not in English, and 852 were not empirical research, leaving 334 articles for further screening stages. Subsequently, three researchers did the title and abstract screening for all 334 articles to determine whether the studies related to the current meta-analysis focus. The title and abstract screening procedures were conducted in independent seatings to ensure that selected studies matched the selection criteria.

3.2.2. Screening of research topic

In the first stage, title and abstract screening was conducted to filter the literature that were not related to the current meta-analysis's focus. The criteria for topic screening were: first, an article that integrated AI agent(s) as an intervention scheme of the experimental group; second, measured one or more student cognitive learning outcomes, using some form of a standardized test.

3.2.3. Screening of experimental studies

In the second stage, literature that only employed a quasi-experimental design was screened. The dependent and independent variables in these studies should be AI agent(s) and student cognitive

learning outcomes, respectively. After screening the literature, the ones that employed either randomized controlled trials (RCTs) or quasi-experimental research designs. Non-experimental studies, literature reviews, and theoretical studies were excluded.

3.2.4. Screening for inclusion or exclusion from the meta-analysis

In the third stage, text screening involved reading each full text individually to identify eligible literature. The goal was to select articles that fully align with the selection criteria. For example, studies needed to have implemented AI agent(s) as an intervention in an experimental group within RCT or quasi-experimental designs, measured student cognitive outcomes, and reported the necessary data for calculating effect sizes, including mean test scores, standard deviation (SD), and sample size N, for both experimental and control groups (Di & Zheng, 2022). Studies that did not measure cognitive learning outcomes, relied on self-reported measures, or had data issues were excluded at this stage. All screening procedures were completed with the agreement of three researchers. After three screening stages, 34 studies were included in this meta-analysis. The application of inclusion and exclusion criteria helped in ensuring methodological rigor in the individual included studies (Liu et al., 2024). The detailed criteria and procedures for screening are summarized in the PRISMA flow diagram (Fig. 1).

Table 1

Descriptive statistics of the studies included in the review.

Study				Participants		Treatment group	Control group	Hedges' g	
Authors	Year	Country	N	Males/ Females	School level	^t reatment	^c ontrol	Effect size	Variance
Feng and Wang	2023	China	85	85 (44/41)	Upper primary school	43	42	0.76	0.06
Zhexenova et al.	2020	Kazakhstan	41	38 (19/19)	Lower primary school	21	20	0.17	0.09
Wongwatkit and Panjaburee	2024	Thailand	336	N/A	Lower secondary school	174	162	0.54	0.01
Chen et al.	2024	China	80	80 (41/39)	Upper primary school	40	40	1.15	0.06
Bachiri et al.	2023	Morocco	100	100 (54/46)	Upper primary school	50	50	0.68	0.04
Chen et al.	2022	Taiwan	60	N/A	Upper primary school	30	30	1.07	0.08
Pareto et al.	2022	Sweden	20	20 (10/10)	Lower and upper primary school	10	10	0.24	0.19
Fang et al.	2025	China	80	80 (48/32)	Upper secondary school	40	40	0.40	0.05
Tong et al.	2025	China	115	N/A	Upper secondary school	55	60	-0.73	0.04
Pei et al.	2025	China	99	N/A	Upper secondary school	49	50	0.45	0.04
Tasdelen and Bodemer	2025	Germany	114	114 (59/55)	Upper primary school	57	57	0.21	0.04
Chen et al.	2025	China	66	66 (30/36)	Lower secondary school	33	33	0.52	0.06
Chen and Liu	2024	China	81	81 (40/41)	Lower secondary school	41	40	0.42	0.05
Ye et al.	2025	China	75	75 (39/36)	Lower secondary school	36	39	0.16	0.05
Liu et al.	2024	China	65	65 (0/65)	Upper primary school	32	33	0.31	0.06
Kwon et al.	2023	Korea	75	75 (38/37)	Upper primary school	37	38	0.18	0.05
Khazanchi et al.	2025	United States	58	58 (32/26)	Lower secondary school	43	15	0.05	0.09
Sapan and Uzun	2024	Turkey	66	66 (28/38)	Upper secondary school	33	33	0.08	0.06
Yang et al.	2025	China	153	N/A	Upper secondary school	76	77	-0.13	0.03
Baha et al.	2024	Morocco	72	72 (38/34)	Upper secondary school	34	38	0.13	0.06
Liu and Chen	2023	China	72	N/A	Upper primary school	36	36	0.45	0.06
Klar	2025	Germany	106	106 (48/54)	Upper secondary school	53	53	-0.09	0.04
Xing et al.	2025	United States	212	N/A	Lower secondary school	107	105	0.01	0.02
Ali et al.	2021	United States	79	79 (39/40)	Lower primary school	34	33	0.63	0.06
Chen and Chang	2024	China	202	202 (103/99)	Lower secondary school	69	70	0.14	0.03
Huang and Qiao	2024	China	136	136 (66/70)	Upper secondary school	68	68	0.64	0.03
Hsu and Hsu	2025	China	56	N/A	Lower secondary school	28	28	0.40	0.07
Elmaadaway et al.	2025	Egypt	90	90 (42/48)	Upper primary school	45	45	2.12	0.09
Liang et al.	2024	China	39	N/A	Upper primary school	17	22	0.08	0.1
Lin and Mubarak	2024	China	52	52 (29/23)	Upper primary school	28	24	0.68	0.08
Mageira et al.	2022	Greece	35	35 (10/25)	Lower primary school	18	17	0.05	0.11
Alneyadi and Wardat	2023	United Arab Emirates	122	122 (64/58)	Upper secondary school	58	64	0.51	0.05
Liu et al.	2025	China	104	104 (60/44)	Lower primary school	52	52	0.09	0.03
Rücker and Becker-Genschow	2025	Germany	195	195 (104/91)	Lower secondary school	102	93	-0.27	0.02

Note: Total sample: 3042. N/A: not available.

3.3. Coding framework

Following the procedures from prior meta-analysis, the included literature were coded following two dimensional categories: descriptive variables and moderating variables (Di & Zheng, 2022; Wang et al., 2024). Descriptive variables captured the basic information of each of the included studies: including the author; year of publication; study context; total number of participants in the study; number of males and female participants; number of participants in the experimental and control groups; the effect sizes, and their corresponding variances in each of the studies. These variables are presented in the descriptive statistics table, in Table 1.

For the potential moderators, data that may influence the effects of AI agents on student cognitive development were extracted. The coding criteria were: 1. Types of AI agents, which were categorized into AI

educational robotics, ITS, GenAI agents, and conversational AI agents. This categorization is based on the framework established by other researchers (Istrate, 2025). 2. Cognitive outcomes categories, which are knowledge-based outcomes (outcomes related to recalling of concepts, principles, such as recalling and writing the Latin letters), skills-based outcomes (outcomes that entail practice and application of specific ability, such as mathematics-problem solving, oral reading fluency), and higher-order thinking skills (outcomes that require analysis, reasoning, and skills beyond recalling of information; for instance scientific thinking, verbal creativity) (Krathwohl, 2002; Wilson & Leslie, 2016). 3. Learner levels, including pre-school (before primary school), lower-primary (grades 1-3), upper-primary (grades 4-6), lower-secondary (grades 7-9), and upper-secondary/high school (grades 10-12) (Wang et al., 2024). 4. Discipline, which is classified according to the instructional content under intervention, includes

Table 2
Characteristics of the intervention for each study included in the review.

Study		Intervention Characteristics					
Authors	Year	Intervention subject	Instructional content	Intervention time: W(S)	Session duration (minutes)	Cognitive outcome category	No. of outcomes
Feng and Wang	2023	Bilingual Education	English/Chinese Skills	24(120)	112	Skill-Based Outcome	3
Zhexenova et al.	2020	Kazakh language	Handwriting instruction	N/A	30	Knowledge-Based Outcome	1
Wongwatkit and Panjaburee	2024	Computer science	Digital literacy concepts, tools, and skills	N/A	N/A	Skill-Based Outcome	1
Chen et al.	2024	Chinese language	Chinese poetry	4	20	Skill-Based Outcome and Higher-Order Thinking	2
Bachiri et al.	2023	Cross-disciplinary digital content	Gamified assessments	3	N/A	Skill-Based Outcome	1
Chen et al.	2022	Art	Color drawing	N/A	N/A	Skill-Based Outcome	1
Pareto et al.	2022	Mathematics	Card-based arithmetic game	1(2)	70	Skill-Based Outcome	1
Fang et al.	2025	Computer science	Basic Python	4 (1)	40	Skill-Based Outcome and Higher-Order Thinking	2
Tong et al.	2025	Physics	Scientific problem-solving in physics	N/A	120	Skill-Based Outcome and Higher-Order Thinking	2
Pei et al.	2025	EFL	EFL writing	2 (13)	40	Skill-Based Outcome	1
Tasdelen and Bodemer	2025	Mathematics	Mathematics problem solving	N/A	60	Skill-Based Outcome	1
Chen et al.	2025	History	Arts-based learning system	8	N/A	Skill-Based Outcome	1
Chen and Liu	2024	Biology	Cell division	2	135	Skill-Based Outcome	1
Ye et al.	2025	Computer science	Python Programming Basics	11 (11)	40	Skill-Based Outcome and Higher-Order Thinking	2
Liu et al.	2024	EFL	EFL writing development	5 (10)	N/A	Skill-Based Outcome	1
Kwon et al.	2023	EFL	EFL writing skills	15 (10)	15	Skill-Based Outcome	1
Khazanchi et al.	2025	Mathematics	Personalized learning	5 (25)	50	Knowledge-Based Outcome	1
Sapan and Uzun	2024	EFL	Paragraph writing in English	3 (6)	40	Knowledge-Based Outcome and Skill-Based Outcome	2
Yang et al.	2025	Computer science	Programming education	6 (12)	N/A	Skill-Based Outcome	1
Baha et al.	2024	Information Technology	Programming language content	N/A	120	Knowledge-Based Outcome	1
Liu and Chen	2023	EFL	Vocabulary learning	11	90	Knowledge-Based Outcome	1
Klar	2025	General education	Content adaptation	N/A	20	Knowledge-Based Outcome	1
Xing et al.	2025	Mathematics	Algebra and Geometry	2 (3)	42	Knowledge-Based Outcome	1
Ali et al.	2021	Creativity	Collaborative drawing activities	N/A	N/A	Higher-Order Thinking	2
Chen and Chang	2024	Science	Newtonian mechanics	2 (10)	60	Skill-Based Outcome	1
Huang and Qiao	2024	Information Technology	Computational thinking	12	40	Higher-Order Thinking	1
Hsu and Hsu	2025	Computational thinking	Generative AI Drawing	N/A	620	Higher-Order Thinking	2
Elmaadaway et al.	2025	EFL	Oral reading fluency	N/A	N/A	Knowledge-Based Outcome and Skill-Based Outcome	4
Liang et al.	2024	Science	Plant diversity	N/A	60	Higher-Order Thinking	1
Lin and Mubarak	2024	Information Technology	Computational thinking	5	40	Skill-Based Outcome	1
Mageira et al.	2022	EFL	Language and content learning	N/A	N/A	Knowledge-Based Outcome	2
Alneyadi and Wardat	2023	Physics	Electromagnetism	12 (3)	N/A	Knowledge-Based Outcome	1
Liu et al.	2025	Mathematics	Mathematical reasoning	N/A	40	Skill-Based Outcome	1
Rücker and Becker-Genschow	2025	Mathematics	Square roots	N/A	N/A	Skill-Based Outcome	1

Note: W = weeks; S = sessions; No. = number; N/A = not available.

natural sciences (biology, chemistry, and physics), mathematics and information technology/computer science, language and literacy, and social sciences (history, arts, creativity, and cross-disciplinary subjects), and others. 5. Instructional content, which specified the content within a specific discipline, was covered under AI agent intervention. 6. Intervention duration reported by weeks in sessions, and minutes, which were classified into less than a month, one to three months, more than three months, and not specified. 6. Number of cognitive outcomes specifically for each of the included studies is reported.

The coded features for the moderator variables are presented in Table 2 (summarizing the intervention characteristics of each study) and Table 5 (Analyzing the moderator variables).

To this end, to ensure the reliability of the coding procedures, three independent coders reviewed all the included studies, and the inter-coder agreement was calculated using the reliability coefficient (R). The formula used was $R = (n \times K) \div (1 + (n - 1) \times K)$ where n is the number of coders and K is the average mutual agreement. Since the average coding agreement between the coders goes by the formula $K_{AB} = 2MA_B \div (NA + NB)$ where MA_B represents the same number coded by two coders, NA and NB are the total numbers of codes by two coders A and B, respectively; assuming the average mutual agreement for the pairs of three coders is K_{AB} , K_{AC} , and K_{BC} , K between three coders is $[K_{AB} + K_{BC} + K_{AC}] / 3$ (Holsti, 1969). Discrepancies in the coding process were discussed among the coders and resolved through consensus before finalizing the coded dataset. The final R was 0.995, indicating good reliability of the coded results (Yeaton & Wortman, 1993).

3.4. Data analysis

To thoroughly examine the influence of AI agents on student cognitive learning outcomes in K-12 classrooms, this study employed a random effect model to estimate the overall effect sizes, assess heterogeneity, check the potential publication bias, and perform moderator variable analyses (Hedges & Olkin, 1985). All the analyses were conducted in STATA 18.0 version.

3.4.1. Calculating ES

Hedge's g was utilized as the effect value to minimize variability in calculations, considering differences in studies' designs and the variation in sample sizes (Hedges et al., 2010). The effect size was computed using the reported primary data from included studies, such as mean test scores, SD, and sample sizes. The effect size was calculated for each study, and when a study has multiple group designs, the effect size was calculated in each group. For empirical studies with more than one cognitive learning outcome, the effect size was calculated for each outcome. The effect sizes are interpreted as small (0.2), moderate (0.5), and large (0.8) (Cohen, 1992).

3.4.2. Homogeneity analysis and moderate analysis

To assess the variation of effect sizes across studies and guide the proper selection of an appropriate effect model based on the heterogeneity results, this study applied the Q-test and I^2 -test. A significant Q-test indicates that the variability of effect sizes across studies is greater than would be expected from sampling errors alone, and thus, suggests heterogeneity in the included studies. The I^2 quantifies proportion of variation in the attributed heterogeneity across studies, with edges of 25%, 50%, and 75% suggesting low, medium, and high variations, respectively (Higgins et al., 2003).

The presence of heterogeneity in the included studies suggests that moderator variables may influence the effectiveness of AI agents on student cognitive development. In this case, subgroup analyses were conducted, and an additional Q-test (Q_B) was conducted to check for the variation in effect sizes across moderator variables.

3.4.3. Investigating publication bias and sensitivity analysis

Publication bias correspond to scenarios where studies report

statistically significant effects having a larger chance of being accepted and published than non-significant ones or with lower effect sizes (Lee, 2025). In this meta-analysis, the funnel plot of effect sizes distribution was used to estimate the potential of publication bias across the included studies.

Since there can be subjectivity in the qualitative judgement of publication bias in the meta-analysis samples, the study employed another approach to overcome such limitations (Greenland, 1994). Specifically, the trim-and-fill method was utilized to estimate the number of potentially missing studies and to correct the overall effect size of the observed asymmetry of the studies' distribution in the funnel plot (Rosenthal, 1979).

In addition to the publication bias analysis, a sensitivity analysis was conducted to examine the outliers that would have an effect on the effect size. One-Study-Removal analysis was performed to determine the influence of the removed as well as the retained studies on the overall effect size, especially for the studies with larger positive or negative effect sizes.

4. Results

4.1. Descriptive information

A total of 34 studies, with 73 effect sizes, were included in this meta-analysis. A combined sample size of 3042 was included; among them, there were 1495 participants in the experimental group, and 1472 people in the control group. The literature search identified the studies published between January 2020 and December 2025, spanning across diverse continental areas, including Asia (67.65%), Europe (14.71%), Africa (8.82%), and North America (8.82%). In terms of participants' distribution, 14.71% involved lower-primary students, 29.41% involved upper-primary students, 2.94% included a mixture of lower and upper-primary students, 26.47% focused on lower-secondary students, and 26.47% examined upper-secondary students. The forms of cognitive categories focused on knowledge-based outcomes (29.41%), skill-based outcomes (58.82%), and higher-order thinking (11.77%). In terms of AI agents, AI educational robots, Intelligent Tutoring Systems, GenAI Agent, and Conversational AI Agent accounted for 21.92%, 13.70%, 23.26%, and 41.12%, respectively. Regarding discipline distribution, language and literacy accounted for 38.36%, mathematics and technology accounted for 38.36%, natural science accounted for 5.48%, arts and creativity accounted for 15.07%, and others accounted for 2.74%. Except for the studies that did not mention the intervention duration (42.47%), most of the experiments were conducted between 1 and 3 months (30.14%), followed by less than one month (19.18%), and more than three months (8.22%).

4.2. Heterogeneity test

The heterogeneity test results of the sample studies are presented in Table 3. The Q statistic is 813.560 ($p < 0.001$), which exceeds the degrees of freedom ($df = 72$), indicating significant heterogeneity among the included studies. The observed I^2 is 91.2%, which is above 75% and reflects differences in effect sizes among included studies. Given this high level of heterogeneity, a random-effects model was employed since it assumes that true effect sizes vary across studies and allows for greater generalizability of findings beyond the sampled dataset (Hedges & Olkin, 1985). Therefore, random-effects model was used to estimate the pooled effect sizes across studies, and moderator analyses were conducted to explore potential sources of heterogeneity.

4.3. Overall effectiveness

The overall effect size of AI agents on student cognitive development in the K-12 setting, with 73 observed effect size values, is 0.404, indicating that AI agents have a moderately significant influence on student

Table 3
Homogeneity test results and combined effect size.

Model	Hedges' g	95% CI	Homogeneity				Tau-squared		
			Q	I ²	df	p	Tau ²	SE	Tau
Fixed	0.269	[0.221, 0.316]	813.560	91.2%	72	<0.001	0.448	0.083	0.669
Random	0.404	[0.242, 0.567]							

cognitive learning outcomes. The 95% CI interval ranges from 0.242 to 0.567, two-tailed test $Z = 4.87$, $p < 0.001$, indicating a statistically significant level attainment, suggesting that AI agents' integrated learning has a statistically significant positive effect on students compared to the conventional methods. Additionally, the overall forest plot illustrating the effect of AI agents on student cognitive learning outcomes is presented in Fig. 2. Out of the observed 73 effects, 13 effect values (17.81%) were classified as large ($g \geq 0.8$), 8 effect sizes (10.95%) were medium ($0.5 \leq g < 0.8$), and 16 effect sizes (21.92%) have small, yet positive values ($0.2 \leq g < 0.5$); while the other 36 (49.32%) had negligible or negative values ($g < 0.2$). This distribution highlights that, while a considerable number of studies showed a moderate to strong positive effect of AI agents intervention, a potential variation remains across contexts.

4.4. Subgroup analysis by cognitive categories

Given that the main focus in the current meta-analysis is the cognitive learning outcomes, the study examined the differential influence of AI agents across three main cognitive outcomes' categories: knowledge-based outcomes, skills-based outcomes, and higher-order thinking (Kratzwohl, 2002).

As shown in Table 4 and Fig. 3, results show that the AI-interventions targeting skills-based outcomes demonstrated a moderate and statistically significant positive effect ($g = 0.391$, 95% CI [0.197, 0.584], $p < 0.001$), suggesting that AI agents are effective in supporting students' development, acquisition, and practice of skills such as problem-solving and computational skills. The significant effect suggests that AI interventions may be well-suited for learning situations where task performance and actual engagement are the central focus. However, substantial heterogeneity was observed across studies ($Q = 409.56$, $I^2 = 90.0\%$), indicating that, while the pooled effect was significant, magnitude of the intervention's effect varies substantially depending on the study's context and instructional features. Similarly, the knowledge-based outcome showed a moderately positive, yet statistically significant effect size ($g = 0.344$, 95% CI [0.040, 0.649], $p = 0.026$). This indicates that AI intervention supports the conceptual understanding and recall of facts, compared to traditional methods. In this regard, heterogeneity in knowledge-based outcomes was high ($Q = 143.82$, $I^2 = 88.2\%$), suggesting a considerable variation in the effect of AI agents on students' basic and conceptual understandings across studies. In terms of higher-order thinking, the effect size was large ($g = 0.540$), but non-significant (95% CI [-0.036, 1.117], $p = 0.066$), indicating that AI agents have a great potential in nurturing higher-order thinking outcomes such as critical thinking skills, analytical reasoning, however, the results are not consistent across the included studies, as the heterogeneity was pronounced in higher-order thinking ($Q = 259.62$, $I^2 = 95.4\%$). Results for cognitive categories are shown in Table 4, and visually displayed in the forest plots in Figs. 3–5. The findings indicate that the observed variation in effect values across studies cannot be explained by sampling error alone, accounting for a moderator analysis to describe which variable(s) may influence these differences.

4.5. Moderator analysis

This study employed a random-effects model approach to determine whether moderator variables influence the effectiveness of AI agents on

student cognitive development and to examine the reasons for variation in effects across the included studies. Results for moderator analysis are presented in Table 5.

The inter-group effect sizes of the AI agents on student cognitive development show a non-significant level ($Q_B = 0.069$, $p = 0.793$), indicating that a specific form of AI agent does not really influence the development of student cognitive learning outcomes in K-12 classrooms. All four types of AI agents demonstrated a positive and significant influence on cognitive development, but with different magnitudes. Influential effect of Intelligent Tutoring Systems is the highest ($g = 0.540$, 95% CI [0.241, 0.839], $p < 0.01$), followed by AI educational robots ($g = 0.474$, 95% CI [0.150, 0.799], $p < 0.001$), Conversational AI Agent ($g = 0.468$, 95% CI [0.008, 0.645], $p < 0.05$), and GenAI Agent ($g = 0.421$, 95% CI [0.198, 0.645], $p < 0.001$).

For the learner level, the strongest effect size was observed in upper-primary level ($g = 0.877$, 95% CI [0.502, 1.253], $p < 0.001$), followed by lower-primary ($g = 0.237$, 95% CI [0.044, 0.431], $p < 0.05$), lower-secondary ($g = 0.195$, 95% CI [0.012, 0.402], $p < 0.05$). However, the effect value was non-significant for mixed lower and upper-primary school students ($g = 0.236$, 95% CI [-0.607, 1.079], $p > 0.05$), or upper-secondary level ($g = 0.037$, 95% CI [-0.217, 0.290], $p > 0.05$). The test for variation in between-group effects shows a significant value ($Q_B = 12.97$, $p = 0.005 < 0.01$). In this case, the results indicate that learner level significantly moderates the influence of AI agents on student cognitive learning outcomes, and AI agents are most effective for younger learners.

For the effect size of each discipline, highest impact was observed in language and literacy ($g = 0.830$, 95% CI [0.384, 1.127], $p < 0.001$), and arts and creativity ($g = 0.755$, 95% CI [0.112, 1.548], $p < 0.05$), followed by mathematics and technology ($g = 0.230$, 95% CI [0.067, 0.393], $p < 0.01$). However, natural science ($g = -0.065$, 95% CI [-0.454, 0.324], $p > 0.05$) and other disciplines ($g = 0.197$, 95% CI [-0.405, 0.798], $p > 0.05$) did not demonstrate significant effects. The test between the homogeneity groups displays that disciplines moderated the effectiveness of AI agents on student cognitive development ($Q_B = 7.61$, $p = 0.006 < 0.01$) significantly, indicating that AI agents are effective in cognitive development across language and literacy, arts and creativity, mathematics and technology, but negligible in natural science and other disciplines.

For the effect size of experimental duration, except for >3 months group ($g = 0.650$, 95% CI [-0.137, 1.436], $p > 0.05$), that demonstrated the largest effect value, but did not reach statistical significant, the rest of durations showed a positive and statistically significant effect; "not mentioned" ($g = 0.463$, 95% CI [0.147, 0.779], $p < 0.01$), 1-3 months ($g = 0.385$, 95% CI [0.185, 0.584], $p < 0.001$), and <1 month ($g = 0.244$, 95% CI [0.031, 0.457], $p < 0.05$). The intergroup test results ($Q_B = 0.23$, $p = 0.632 > 0.05$) indicate that differences in experimental duration did not significantly moderate the effect of AI agent on student cognitive learning outcomes in K-12 education.

4.6. Publication bias test

To assess the robustness of findings against publication bias, this study employed various methods, including the Classic fail-safe N test, Orwin's fail-safe N test, and the funnel plot that distributes effect sizes across studies. The funnel plot (Fig. 6) of standard error against effect size shows that most studies were evenly distributed on both sides of the

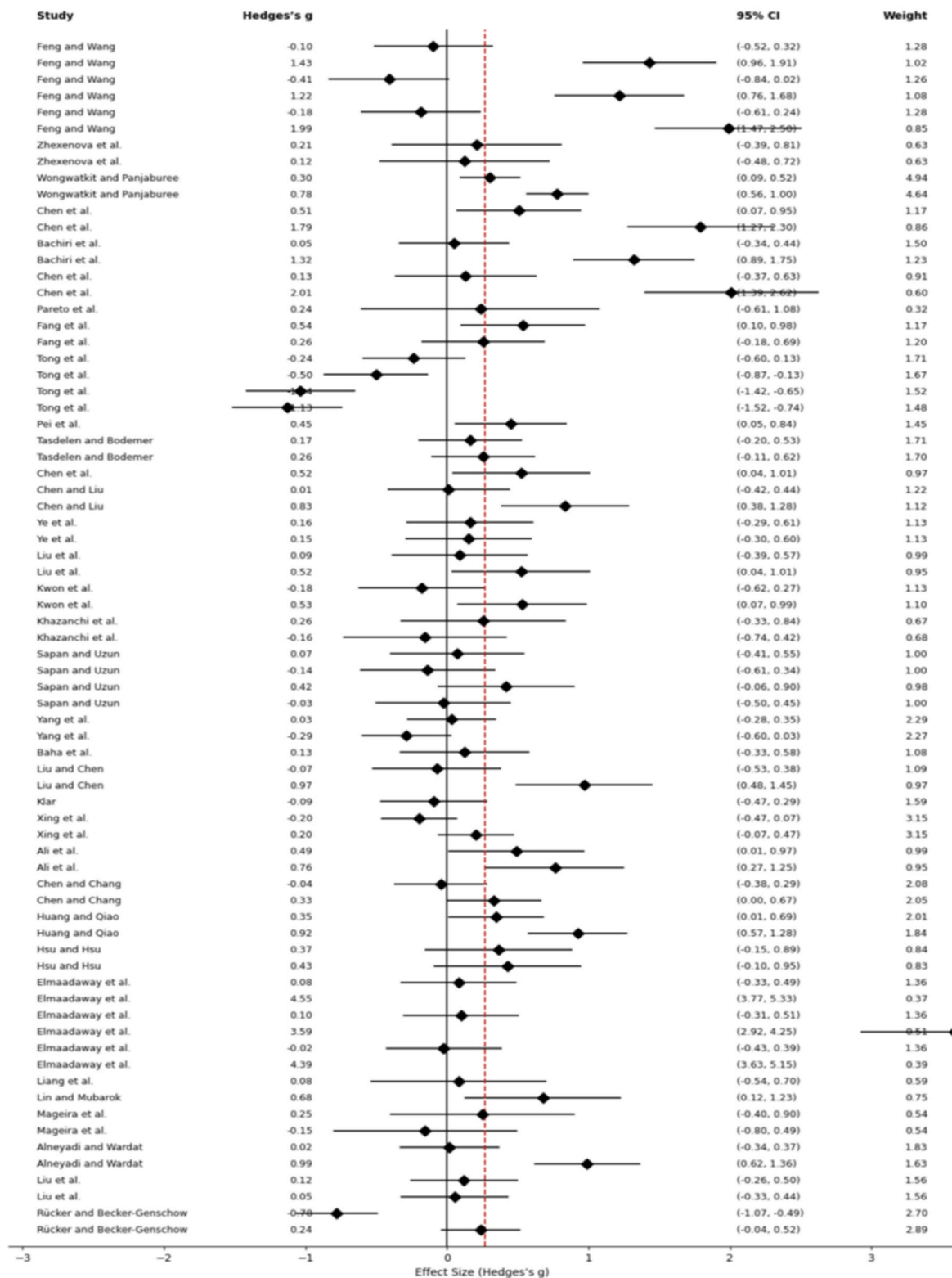


Fig. 2. Forest plot of selected studies.

axis of symmetry; however, a few studies on the right side appear to have relatively larger effect sizes. This indicates that, while most of the studies are centered around the mean, there might be a slight asymmetry; however, this asymmetry is not pronounced enough to suggest publication bias. Even though a slight asymmetry was observed in the funnel plot, the trim-and-fill method was run, and there were no missing

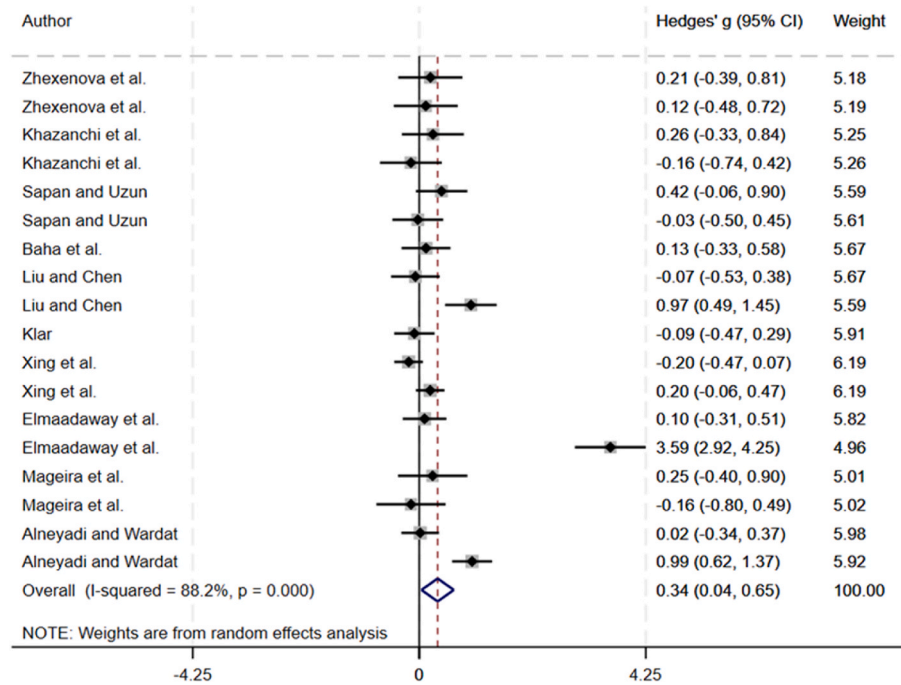
studies that would be required to make the funnel plot symmetrical (Duval & Tweedie, 2000).

The results of the Classic fail N-safe test, presented in Table 6, indicate that a total of 378 studies, slightly more than twice the 180 required ($5k + 10$, where k represents the 34 studies included in this meta-analysis), are necessary to reduce the overall effect size (Rosenthal,

Table 4

Subgroup analysis by cognitive category.

Cognitive Category	k	Hedges' g	95% CI	Q	I ² (%)	Z	P	Tau ²	SE
Knowledge-Based Outcome	18	0.344	[0.040, 0.649]	143.82	88.2%	2.22	0.026	0.370	0.155
Skill-Based Outcome	42	0.391	[0.197, 0.584]	409.56	90.0%	3.95	0.000	0.359	0.099
Higher-Order Thinking	13	0.540	[-0.036, 1.117]	259.62	95.4%	1.838	0.066	1.063	0.294

**Fig. 3.** Forest plot of AI agents on knowledge-based outcome.

1979). In this case, the effect size of unpublished studies does not have a significant influence on the overall effect of the current study. Orwin's fail-safe N results (Table 7) indicate that 2876 unpublished studies would be required to reduce the overall pooled effect size to below the threshold of 0.01.

4.7. Sensitivity analysis

Sensitivity analysis was conducted to examine whether the overall effect size is influenced by any single study included in the meta-analysis. A leave-one-out procedure indicated that the pooled effect size remained stable after the removal of individual studies. Specifically, under the fixed-effect model, the pooled Hedge's g ranged from 0.221 to 0.316, while under the random-effects model, the pooled Hedge's g ranged from 0.242 to 0.567. These results highlight that no single study substantially altered the magnitude or direction of the overall effect, supporting the robustness of the meta-analytic findings.

5. Discussion

Several studies have documented the role of AI in K-12 settings (Azzam & Charles, 2024; Crompton & Burke, 2022; Yi et al., 2025). However, there is a contradiction regarding how this particular technology impacts students' learning, and in many cases, the studies did not specifically consider the development of cognitive learning outcomes. To address these inconsistencies and the existing gap, this study synthesizes the empirical literature. It adopts a random-effects model to examine the overall effectiveness of AI agents on student cognitive learning outcomes in K-12 education and to investigate potential moderators that may influence this effectiveness. A total of 73 effects drawn

from 34 (quasi) experimental studies with 3042 participants were analyzed.

5.1. How effective is an AI agent in enhancing student cognitive learning outcomes in a K-12 setting?

The results show that AI agents have a moderately positive effect on student cognitive learning outcomes, with an overall effect size of ($g = 0.404, p < 0.001$). This suggests that AI agents are not merely tools to substitute traditional teaching, but act as active mediators of learning, especially in the 21st-century era. Basically, cognitive learning entails students' active engagement, adaptability, and interaction (Elim, 2024; Krathwohl, 2002). Unlike static digital technology, AI acts to respond to learners' inserted information, allowing for personalized scaffolding, providing interactive learning environments, and offering immediate feedback that enhances knowledge and skills development (Huang & Qiao, 2024; Wei et al., 2025; Zhou et al., 2024). For instance, conversational AI not only provides a direct answer to students' questions but also offers progressive steps to approach the solution and guides students to articulate reasoning, thereby fostering cognitive monitoring (Yusuf et al., 2025; Hsu & Hsu, 2025). AI agents can stimulate real task scenarios and motivate learners to practice repetitively (Liu, Guo, & Hu, 2025; Wang et al., 2025). In the K-12 setting, where learners build foundational knowledge and skills, the kind of repeated interaction with learning tools and active engagement is crucial (Crompton & Burke, 2022). For instance, Intelligent Tutoring Systems and adaptive AI allow younger learners to practice their learning, such as writing, speaking, and arithmetic skills, until mastery is attained (Elmaadaway et al., 2025; Xing et al., 2025). Generative AI has similarly been found to produce problem-solving scenarios and offer immediate feedback, where

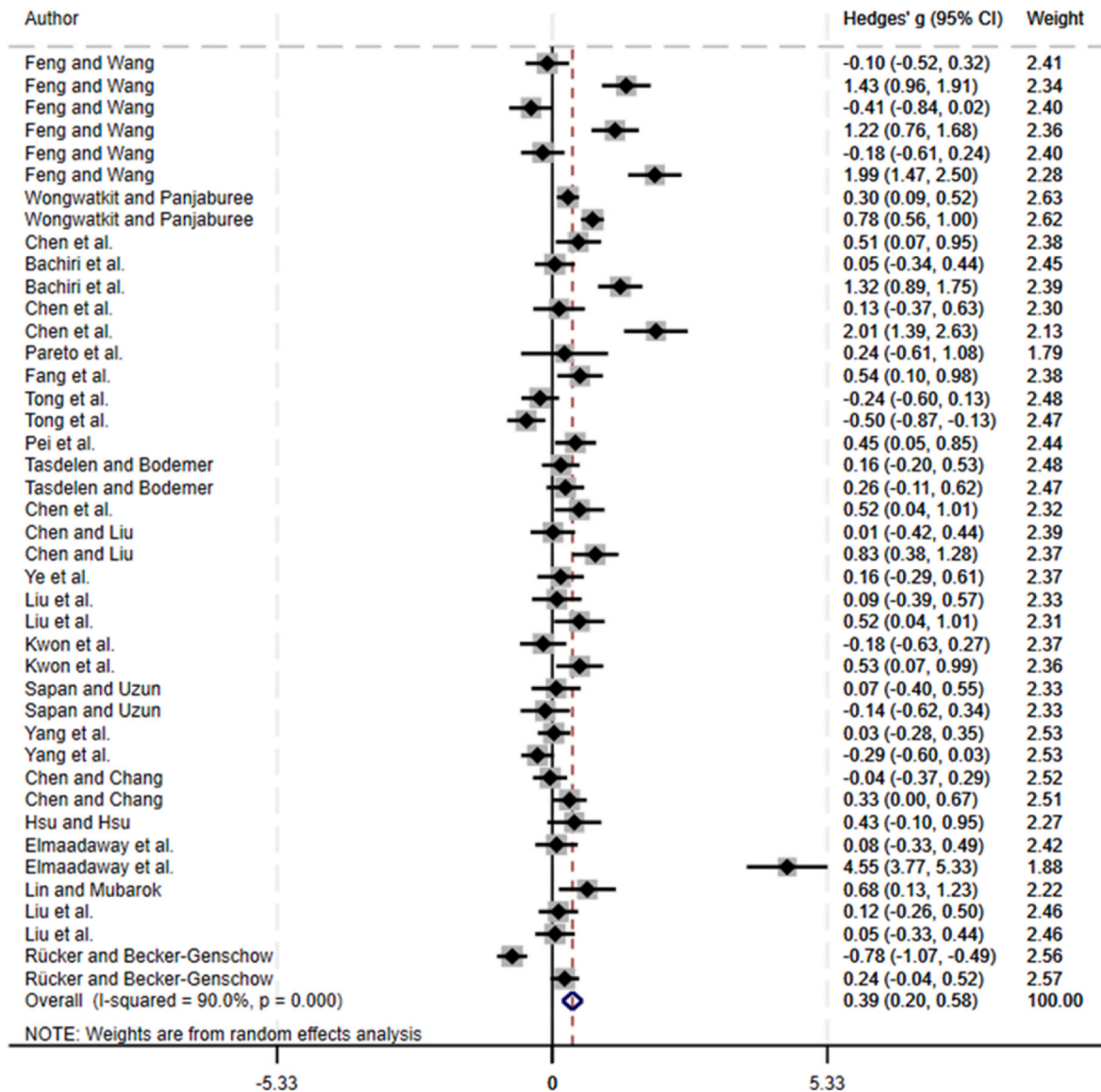


Fig. 4. Forest plot of AI agents on skills-based outcome.

students get an opportunity to rehearse strategies and develop skills for problem-solving tasks (Gökoğlu & Erdoğan, 2025). These findings align with some previous studies that reported a statistically significant effect of AI technology on student cognitive development (Vieriu & Petrea, 2025; Yi et al., 2025). All these features support the development of cognitive learning outcomes by enhancing the acquisition of basic skills while fostering the achievement of higher-order thinking in appropriate means.

5.2. What is the effectiveness of AI agents across differential categories of cognitive learning outcomes?

The subgroup analysis revealed that the influence of AI-agents is not uniform on students' learning outcomes in K-12 classrooms; it varies according to the targeted cognitive skills. A specific analysis of the cognitive outcomes' categories revealed that AI agents exerted a statistically significant moderate effect on skills-based outcomes ($g = 0.391$, $p < 0.001$). This suggests that AI-driven interventions are tailored to enhance engagement, scaffold strategies, and guide problem-solving skills (Kwon et al., 2023; Mageira et al., 2022). Since the K-12 learners are still developing cognitively, they benefit most through

procedural learning, which is fostered through adaptive, reinforcement learning, and responsive feedback (Holmes et al., 2019). Therefore, by integrating the AI agents with instructional approaches that emphasize guided problem-solving tasks, interactive learning, and adaptive scaffolding, schools can nurture students' competencies and skills in domains such as vocabulary, computational thinking, and oral fluency (Feng & Wang, 2023; Tong et al., 2025). The small and significant effect of AI agents on knowledge-based outcomes ($g = 0.344$, $p < 0.05$) indicates that, while AI technology supports factual and conceptual understanding of basic knowledge, its advantages over traditional approaches remain modest (Maamor et al., 2025). Literature points out that teachers and textbooks remain the main sources of knowledge for younger learners (Cheah et al., 2023; Hillmayr et al., 2020). In this case, basic understandings and knowledge gains in K-12 settings are less pronounced with AI technology, especially when the AI tool is integrated with less interactive instructional practices compared to the other cognitive domains. Conversely, a larger effect size was observed in higher-order thinking, but it was not statistically significant ($g = 0.540$, $p = 0.066$). This inconsistency highlights both the promising influences and the challenges of incorporating AI agents in supporting the development of higher-cognitive skills, such as critical thinking and creative

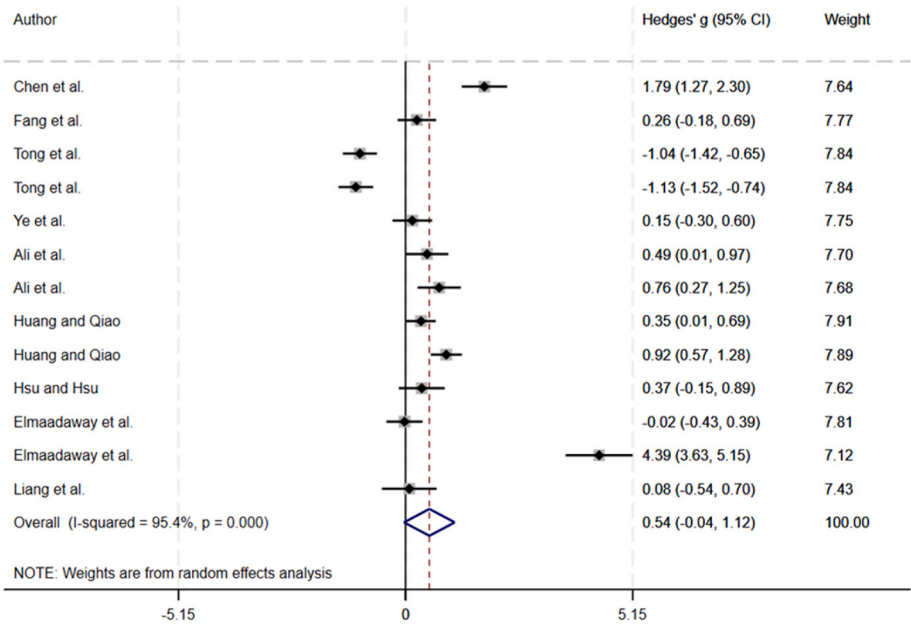


Fig. 5. Forest plot of AI agents on higher-order thinking.

reasoning. When integrated with inquiry-based learning or project-driven classes, AI technology fosters the development of higher cognition in learners (Du et al., 2025; Song et al., 2025). However, its effectiveness depends on key factors such as the embedded instructional design, assessment practices employed within a specific instructional practice (Hu et al., 2025; Liu, Guo, He, et al., 2025). The heterogeneity across the three cognitive categories underscores the need for contextual features, such as learners' level, discipline under intervention, type of AI agent utilized, and the duration of intervention. Considering all these moderating factors, it can be concluded that AI agents are effective in enhancing procedural skills, moderately support factual knowledge acquisition, but show inconsistent potential in fostering higher-order cognitive skills, indicating the need for different designs and future research in this area.

5.3. How do the variables, such as type of AI agent, learner level, disciplines, and experimental duration, moderate the effect?

The observed heterogeneity indicates the significant variability of each of the included studies in this meta-analysis, and thereby necessitates the need to explain this particular variability through moderating effect analysis. To comprehensively understand the influence mechanism of AI agents on student cognitive learning outcomes, this study explores four moderating variables that together describe the reasons for the differences in effect values among the studies.

From the type of AI agent perspective, there were meaningful differences in the influence of specific AI agent types on facilitating the development of student cognitive learning outcomes. Intelligent Tutoring Systems and AI-educational robots demonstrated the most significant effect. This could be because ITS and robotics AI are often human-like teaching technologies, and characterized by providing scaffolding and personalized learning conditions, thereby strengthening the procedural and conceptual knowledge and skills in K-12 settings (Yi et al., 2025). In some cases, since younger learners are still developing cognitive awareness, the types of AI-technology that depend on self-exploration and regulation, such as generative AI agents, may appear to have less significance on their own (Bachiri et al., 2023). These findings conform to those of Li and Wilson (2025), who reported that generative AI tools were found to be effective only when integrated with the scaffolding frameworks, including a teacher's guide and structured

feedback templates. Similarly, while conversational AI agents can enhance students' motivation and engagement, these can contribute to effective cognitive development in learning when embedded with a deliberate instructional design, particularly in a K-12 setting where learners depend on the guidance to convert engagement into meaningful learning (Wang et al., 2025).

From the learner level perspective, a clear pattern emerged. The largest effect was observed in the upper-primary school, suggesting that AI technology may be effective in transitioning the bridge of basic to complex cognitive skills. This finding echoes with the importance of adaptive feedback and guided interactivity in learning, such that they strongly benefit students in middle to upper-primary education (Vygotsky, 1978). Notwithstanding, the small and statistically significant effect observed in the lower-secondary level suggests that, at this level, learners are starting to develop abstract reasoning, and they encounter tasks that demand guidance (Dale, 1970). In this case, AI agents, especially those designed to scaffold reasoning can smooth the development of cognitive skills in these particular learners (Soini et al., 2023). Furthermore, the effect value of AI agents on cognitive outcomes in the upper secondary level was negligible and non-significant. This indicates that learners at this level are transitioning to higher reasoning domains, and thus, they may require more sophisticated AI tools to support their learning (McLeod, 2025). In fact, since learners at the upper-secondary level are preparing for their final examinations and future careers in life, the scaffolding and procedural focus of AI technology may not benefit them (Soini et al., 2023); therefore, attention should be placed on self-regulated learning approaches to facilitate the development of deeper cognitive skills (Cloux et al., 2025). The small effect observed in both the lower-primary and mixed (lower and upper-primary) levels may be attributed to the limited autonomy of young learners in self-monitored interactions with AI tools (Wang et al., 2024). These results highlight the importance of considering developmental stages across education levels when tailoring AI interventions.

The disciplinary differences were found to moderate the influence of AI agents on student cognitive outcomes. A significantly larger effect was demonstrated in language and literacy, as well as in the arts and creativity. This finding underscores the potential features of interactive AI agents with disciplines that rely on creativity, expression, and scaffolding of key competencies such as vocabulary, learning, reading, and writing (Sapan & Uzun, 2024). In K-12 classrooms where language

Table 5

The effect sizes of categories and their related moderator variables.

Moderator	k	g	z	95% CI	Q _B	P value
AI agent					0.069	0.793
AI educational robots	16	0.474	2.86**	[0.150, 0.799]		
Intelligent Tutoring Systems	10	0.540	3.541***	[0.241, 0.839]		
GenAI Agent	17	0.421	3.69***	[0.198, 0.645]		
Conversational AI Agent	30	0.468	2.01*	[0.008, 0.645]		
Learner level					12.97	0.005
Lower-primary school	8	0.237	2.41*	[0.044, 0.431]		
Upper-primary school	28	0.877	4.58***	[0.502, 1.253]		
Lower-secondary school	17	0.195	1.85*	[0.012, 0.402]		
Lower and upper-primary school	1	0.236	0.55	[-0.607, 1.079]		
Upper-secondary school	19	0.037	0.28	[-0.217, 0.290]		
Disciplines					7.61**	0.006
Language and Literacy	28	0.830	3.98***	[0.384, 1.127]		
Mathematics and Technology	28	0.230	2.77**	[0.067, 0.393]		
Natural science	11	-0.065	0.33	[-0.454, 0.324]		
Arts and Creativity	4	0.755	2.27*	[0.112, 1.548]		
Others	2	0.197	0.64	[-0.405, 0.798]		
Experimental duration					0.23	0.632
<1 month	14	0.244	2.24*	[0.031, 0.457]		
1-3 months	22	0.385	3.78***	[0.185, 0.584]		
>3 months	6	0.650	1.62	[-0.137, 1.436]		
Not mentioned	31	0.463	2.87**	[0.147, 0.779]		

Note k = number of included studies; g = mean effect size; CI = confidence interval; Q_B = between-group homogeneity. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

literacy and creativity disciplines focus on dialogues, basic expressions, and vocabulary (Jackson, 2020), AI agents, such as AI educational robots, conversational AI, act to design the collaborative learning activities and offer immediate corrective feedback, thereby fostering reading fluency, oral practice in ways that reflect the meaningful learning process (Azzam & Charles, 2024; Zhexenova et al., 2020). In contrast, the negative and non-significant effect observed in the natural sciences is theoretically informative. Natural science learning typically entails conceptual reasoning, inquiry-based learning, hypothesis testing, and experimentation (Almasri, 2024). Meanwhile, many of the AI agents implemented in K-12 education are primarily designed to provide procedural learning support, step-by-step problem-solving guidance, and automated feedback, which may be insufficient for supporting the development of open-ended inquiry and conceptual exploration required in science learning (Crompton & Burke, 2022). As a result, AI agents in the K-12 context may constrain learners from developing or acquiring authentic scientific inquiry when applied in natural science contexts. This particular finding suggests that the effectiveness of AI agents is domain-dependent, highlighting the need for future AI designs that support inquiry-based learning in natural science contexts.

Finally, in the experimental duration, AI agents did not show a significant difference in cognitive outcomes across intervention durations. However, comparing the effect values of experimental time, the longer duration demonstrated a larger but non-significant effect; shorter-term interventions of less than one month, and medium-term interventions

Table 6

Results of the classic fail-safe N.

The z value for observed studies	4.870
The p-value for observed studies	0.000
Alpha	0.05
Tail	2.00
Z for alpha	1.96
Number of observed studies	73.00
Number of missing studies that would bring the p-value to > alpha	378.00

Table 7

Results of Orwin's fail-safe N.

Hedges' g in observed studies (random effect)	0.404
The criterion for a 'trivial' Hedges' g	0.01
Mean Hedges' g in missing studies	0.00
Number of missing studies needed to bring Hedges' g to under 0.01	2876.00

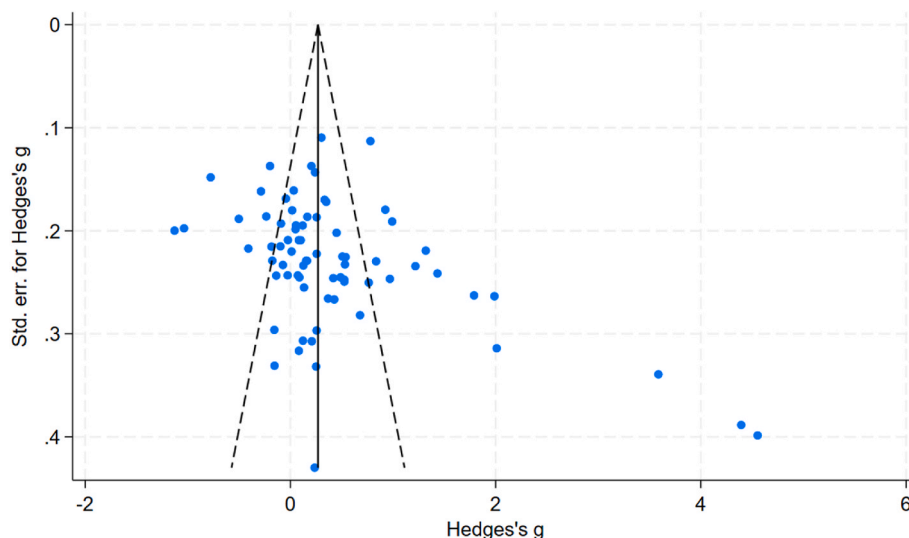


Fig. 6. The funnel plot by Hedge's g (random effects).

of one to three months demonstrated positive and significant effects. This suggests that, while AI agents provide an interactive learning environment and offer an engagement opportunity, their impact may remain superficial when learners are not given sufficient time to practice and internalize the acquired skills (Apedoe et al., 2008). This is consistent with the findings of Gu and Yan (2025) that highlighted the positive linear relationship of AI-intervention duration in facilitating students' learning. The variation in longer-term interventions could be associated with methodological and practical challenges. For instance, with a few included studies in this meta-analysis whose experimental duration extended three months, the statistical power to detect the effect (Valentine et al., 2010). Importantly, implementing AI-interventions in K-12 settings over a long period of time may pose some challenges, such as technological limitations that may further hinder continuous engagement for learners (Holmes et al., 2019). Therefore, the moderator analysis results highlight the note that AI is not a one-size-fits-all tool, but its integration is influenced by factors such as type of AI agent, education level, instructional subject, and intervention duration.

6. Conclusions and implications

6.1. Conclusions

This study employed a meta-analysis to examine the effectiveness of AI agents on students' learning outcomes in K-12 education. Analyzing 34 empirical studies and 73 observed effects, the findings reveal that AI agents have a statistically significant moderate effect on student cognitive learning outcomes. Subgroup analysis shows significant variation, with AI agents having larger effects on skills-based outcomes and higher-order thinking than on knowledge-based outcomes. Moderator variable analysis indicates that the observed effect varies by learner level and discipline. These results affirm the positive influence of AI agents on student cognitive outcomes in K-12 settings, while highlighting contextual and practical features that may affect this relationship.

6.2. Implications of theory and practice

The current study offers useful theoretical and practical insights into AI technology and its implications in learning. From the theoretical perspective, this study adds to a growing body of literature on AI in education by clarifying and confirming the potential effectiveness of AI interventions on cognitive achievements, particularly in K-12 education. Similarly, the findings extend to the existing theoretical frameworks (e.g., Vygotskian scaffolding and constructivist learning frameworks) by highlighting that AI agents can act as cognitive facilitators that personalize instructions, scaffold the learning process, and offer immediate corrective responses across K-12 contexts. From a practical perspective, the moderating analysis results of this meta-analysis point out that contextual features, such as learner level and discipline under intervention, should be considered when integrating AI technology in K-12 education. In that case, future research should put a focus on the following key points: (1) educators should prioritize implementing the AI-interventions at the upper-primary education level, because in this window, learners are transitioning to abstract reasoning and problem-solving stages; therefore, AI technology is essential. (2) instructors should be strategically aligning AI-intervention with subject-specific required output. For instance, leverage AI agents designed for interaction, such as chatbots for vocabulary practice; GenAI for task design, ITS for procedural guidance in mathematics, so as to enhance wide cognitive development across various domains. (3) schools should design AI-supported programs with a balance of sustainability and intensity, and future research may wish to explore how long-term AI interventions can foster cognitive gains.

6.3. Limitations of the study

Several limitations warrant additional note: First, although the included studies cover multiple countries and education levels within K-12 settings, the uneven distribution of studies across variables such as AI agent types and disciplines limited some moderator analyses. Second, moderator analysis was constrained by the insufficient primary data; potentially influential factors, such as instructional strategies and teacher roles, were often unavailable. Third, although publication bias was examined using several statistical approaches (e.g., Classic fail-safe N test, Orwin's fail-safe N test, trim-and-fill method, and funnel plot), the current study did not conduct a formal methodological risk-of-bias or quality appraisal of individual primary studies using the standardized assessment tools. In that case, the internal validity of the primary studies could not be systematically evaluated, and the pooled results should be construed with caution. Therefore, future meta-analyses should incorporate the formal methodological quality assessment tools to strengthen causal inference and internal validity of the findings.

Fourth, the majority of included studies (67.65%) originated from Asian educational contexts, limiting the global generalizability of the findings to regions with different curricular structures, technological infrastructures, and practices. Lastly, this review includes only the experimental and quasi-experimental studies with sufficient data. Incorporating qualitative results could provide more detailed insights into how AI agents influence cognitive achievement. Future research should consider combining meta-analysis with qualitative methods to gain a comprehensive understanding of the effect and dynamics of AI interventions in K-12 context.

Declaration of the use of generative AI

The authors declare that no generative artificial intelligence or any other AI-assisted technology was used in the data preparation, analysis, writing of the manuscript, or interpretation of findings.

CRediT authorship contribution statement

Ji Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Airini Erasmi Mbowe:** Writing – review & editing, Writing – original draft, Formal analysis. **Dahman Tahri:** Writing – review & editing. **Millicent Aziku:** Writing – review & editing.

Ethics approval and consent to participate

Ethical approval was not required for this study as it did not involve human or animal subjects.

Consent for publication

All authors have approved the final version of the manuscript and consent to its publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Summary of the current literature review of AI in K-12 education

Reference	Study nature	Objective	Findings
Lee and Kwon (2024a)	Meta-analysis	Synthesizing the effectiveness of AI education in K-12 Korean classrooms	AI exhibited a larger effect size in cognitive development ($ES = 1.060$) than in the affective domain ($ES = 0.597$). The authors point out that the current education system in Korea insists on the integration of AI technology in enhancing the core cognitive skills of students, such as applications and algorithms.
Azzam and Charles (2024)	Systematic review	Analyzing the overall implementation of AI in K-12 education	Among other roles, AI is essential in facilitating students' personalized learning experiences.
Liu and Zhong (2024)	Systematic review	Developing AI education in the K-12 context	With 45 empirical studies reviewed, the study concludes that AI in K-12 education has the potential to develop students' knowledge, affectivity, and thinking skills.
Yi et al. (2025)	Systematic review and meta-analysis	Investigating the effectiveness of AI in improving mathematics performance in the K-12 classroom	The overall results indicate a significant positive effect on the impact (0.343); however, the two AI agents (AI-intelligent tutoring system and adaptive learning systems) demonstrated a greater impact on students' mathematics performance.
Liu et al. (2025)	Meta-analysis	Examining the impact of GenAI on the students' learning outcomes in K-12 and higher education	The analyzed 49 empirical studies resulted in positive, significant effect sizes; learning achievement (0.857), which was quite greater than learning motivation (0.803). The study indicates that the observed effect size varied according to moderators such as level of education, discipline under intervention, intervention duration, and the instructional technique employed under AI-intervention.
Létourneau et al. (2025)	Systematic review	Investigating the effect of Intelligent Tutoring Systems (ITSs) on K-12 students' learning and performance	The 28 analyzed studies with a total of 4597 students came out with the conclusion that ITSs are efficient in supporting K-12 students' learning, and hence aid in lifting their performances.
Deng et al. (2025)	Systematic review and meta-analysis	Examining the impact of ChatGPT on students' learning	Upon synthesizing 69 articles, the findings reveal that ChatGPT improves students' academic performance, particularly the learning gain, higher thinking skills, as well as learning motivation.
Palioura and Sapounidis (2025)	Scoping review	To map the conducted research on the effects of storytelling and robotics in education	The findings show that the AI tool has no significant impact on self-efficacy. The interventions (storytelling and robotics) resulted in students' development of skills in various domains, including cognitive and affective ones. However, it is reported that the impact of the two technologies varies per learners' education level and disciplines.
Zhu et al. (2025)	Meta-analysis	Quantifying the impact of GenAI on students' learning outcomes	GenAI demonstrated a smaller, yet significant effect on students' overall learning outcomes. The study further extends that, with a longer-term support of GenAI, students can improve their learning achievement.
Lee and Kwon (2024b)	Systematic review	Overview of AI education in the K-12 context	The findings highlight that, among others, AI supported the development of students' problem-solving skills and AI literacy.
Istrate (2025)	Scoping review	Examining the impact of educational AI agents on student engagement and learning outcomes	AI agents tend to increase students' motivation through personalized learning and immediate feedback. This tendency tends to significantly improve their learning outcomes.

Data availability

Data is collected from public sources, and can be made available upon reasonable request to the corresponding author.

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